



Batterjee Medical College
For Science and Technology

Cardiovascular Risk Reduction

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Overview

- Lifestyle Modification
- Diet
- Physical Activity
- Hypertension
- Hyperlipidemia
- Diabetes Mellitus
- Aspirin

Learning Objectives

Knowledge & understanding

Classify the modifiable and non-modifiable risk factors for atherosclerotic cardiovascular disease (ASCVD).

Describe the 10-year ASCVD risk categories and their role in guiding preventive therapy.

Skill

Calculate a patient's 10-year ASCVD risk score to guide decisions on statin and aspirin therapy.

Formulate an individualized, evidence-based cardiovascular risk-reduction plan integrating lifestyle modification and pharmacotherapy.

- Overview and Epidemiology
(global and Saudi burden of cardiovascular disease, morbidity trends)
- Traditional ASCVD Risk Factors (modifiable and non-modifiable)
- 10-year ASCVD risk categories
- Lifestyle Modification:
 - Diet
 - Physical Activity
 - Tobacco Cessation
 - Hypertension
 - Hyperlipidemia (
 - Diabetes Mellitus
 - Aspirin for Primary Prevention

- How many of your relatives or family have had a heart attack or stroke?
- How has that illness affected their lives?



- Coronary heart disease is associated with more than **2.5 million hospitalizations** worldwide each year.
- It is estimated that a myocardial infarction occurs every **34 seconds** in the US, and that one **person dies each minute** from a major coronary event.

Overview

- Cardiovascular disease is a leading cause of mortality globally
- It includes hypertension, coronary artery disease, heart failure, congenital defects and strokes

Total deaths in the United States from COVID-19 and other leading causes, 2020-2022

	Category	Total deaths (Jan.-Sept. 2022)	Total deaths (2021)	Total deaths (2020)
1	Heart disease	572,336	767,937	764,512
2	Cancer	454,176	604,358	599,607
3	COVID-19	234,434	475,059	343,566
4	Accidents	170,166	226,987	203,033
5	Stroke	123,215	162,769	159,248
6	Chronic respiratory	107,559	141,906	152,051
7	Alzheimer	87,866	119,442	134,271
8	Diabetes	74,716	103,197	101,355
9	Other respiratory	50,635	66,381	66,053
10	Renal failure	42,596	53,057	51,221

Notes: For 2022, the total death sum for each category is for January 1 - September 30, 2022, except deaths from accidents and suicides are from January - September 2021. Chronic respiratory is chronic lower respiratory disease.

Source: KFF analysis of CDC mortality and KFF COVID-19 tracker data

Morbidity Rates

MI death rates have declined over past several decades due to

- Heightened public awareness
- Increased availability of automated external defibrillators (AED)
- Improved cardiovascular diagnosis and therapy
- Use of cardiovascular drugs by persons at high risk
- Improved revascularization techniques
- Improved, more aggressive risk factor modification

Traditional ASCVD risk factors

Nonmodifiable risk factors

- Advancing age
- Male sex
- Race and ethnicity

Modifiable risk factors

- Smoking
- Diabetes mellitus
- Hypertension
- Dyslipidemia (↑ total cholesterol, ↓ HDL cholesterol)

10-year ASCVD risk categories

10-year ASCVD risk categories

The 10-year risk categories estimate the risk of developing **myocardial infarction** or **stroke** in this time period and are used for individuals aged 40–75 years.

- Low risk: $< 5\%$
- Borderline risk: 5–7.4%
- Intermediate risk: 7.5–19.9%
- High risk: $\geq 20\%$

- https://www.mdcalc.com/calc/3398/ascvd-atherosclerotic-cardiovascular-disease-2013-risk-calculator-aha-acc?uuid=85cb2533-085a-4ae3-b953-7357cde73d82&utm_source=mdcalc



AHA Life's Simple 7

Life's Simple 7

1. Get Active
2. Eat Better
3. Lose Weight
4. Stop Smoking
5. Control Cholesterol
6. Manage Blood Pressure
7. Reduce Blood Sugar

1. ≥ 150 minutes moderate activity /week or ≥ 75 minutes vigorous activity/week
2. Eat a healthy diet (4-5 components of healthy diet score*)
2. Have a normal body weight (BMI < 25)
3. Never smoked or quit >1 year ago
4. Total cholesterol <200 mg/dL
5. Blood pressure <120/80 mm Hg
6. Fasting blood glucose <100 mg/dL

* 1) 4.5 cups or more of fruits and vegetables per day 2) two or more 3.5-oz servings of fish per week 3) three servings per day of whole grains 4) less than 1500 mg of sodium per day 5) 36 ounces or less of sugar-sweetened beverages per week

Recommended diet intakes

- Saturated fatty acids to account for <10% of total energy intake, through replacement by polyunsaturated fatty acids.
- Trans unsaturated fatty acids <1% of total energy intake.
- <5 g of salt per day.
- 30–45 g of fibre per day, from wholegrain products, fruits and vegetables.
- 200 g of fruit per day (2–3 servings).
- 200 g of vegetables per day (2–3 servings).
- Fish at least twice a week, one being oily fish.

Recommended diet intakes

- Numerous observational studies have demonstrated the benefit of a Mediterranean style diet on cardiovascular risk reduction

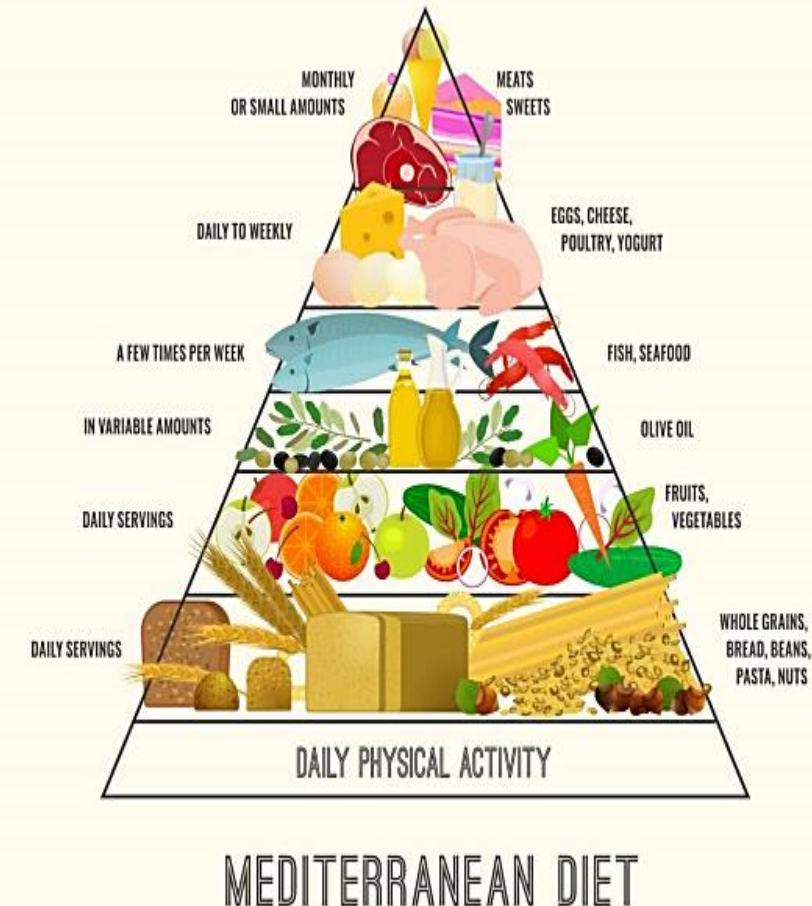
THE NEW ENGLAND JOURNAL of MEDICINE

ORIGINAL ARTICLE

Primary Prevention of Cardiovascular Disease with a Mediterranean Diet Supplemented with Extra-Virgin Olive Oil or Nuts

R. Estruch, E. Ros, J. Salas-Salvadó, M.-I. Covas, D. Corella, F. Arós, E. Gómez-Gracia, V. Ruiz-Gutiérrez, M. Fiol, J. Lapetra, R.M. Lamuela-Raventós, L. Serra-Majem, X. Pintó, J. Basora, M.A. Muñoz, J.V. Sorlí, J.A. Martínez, M. Fitó, A. Gea, M.A. Hernán, and M.A. Martínez-González, for the PREDIMED Study Investigators*

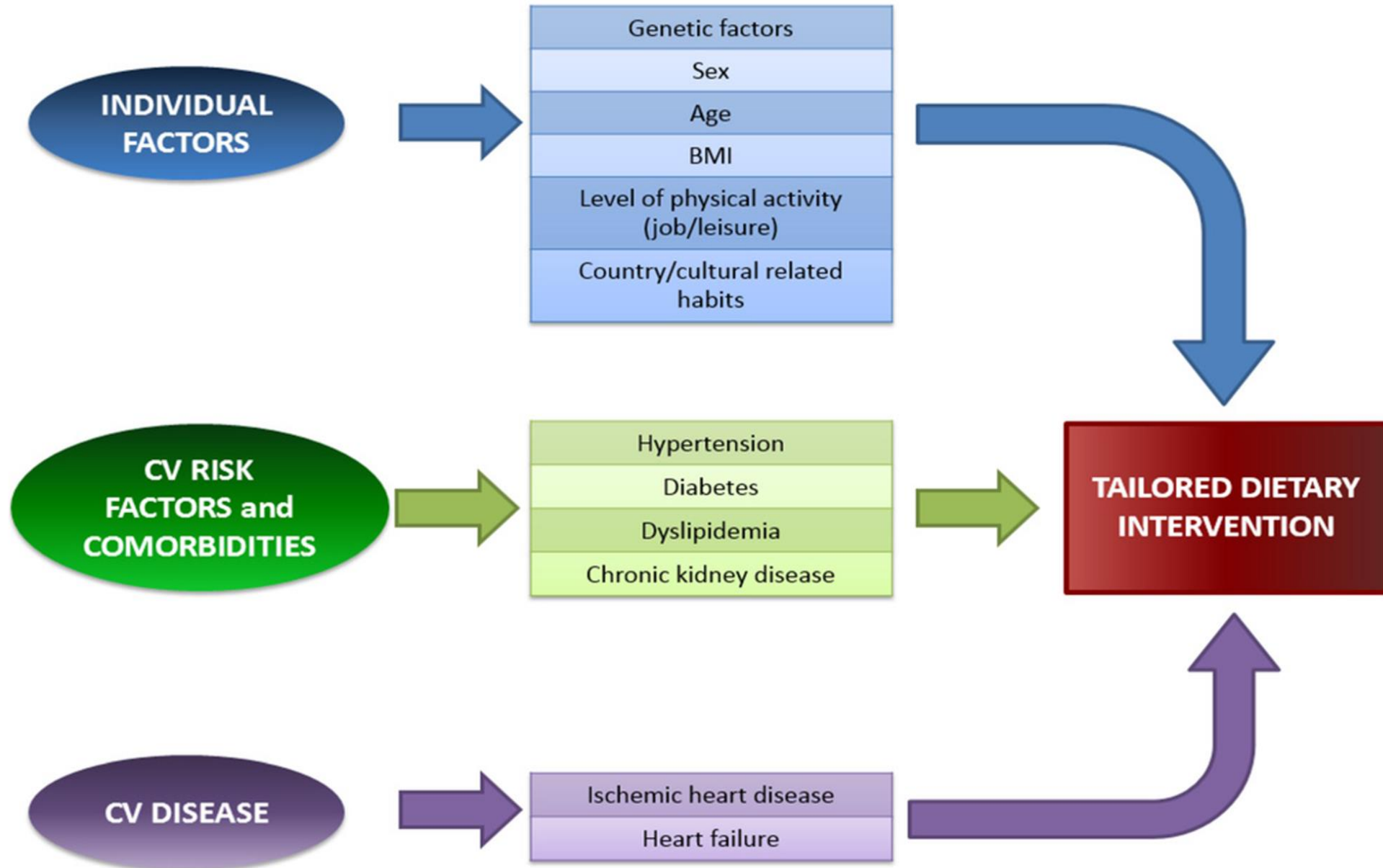
N Engl J Med 2018; 378:e34
DOI: 10.1056/NEJMoa1800389



Nutritional counseling



Recommended diet intakes

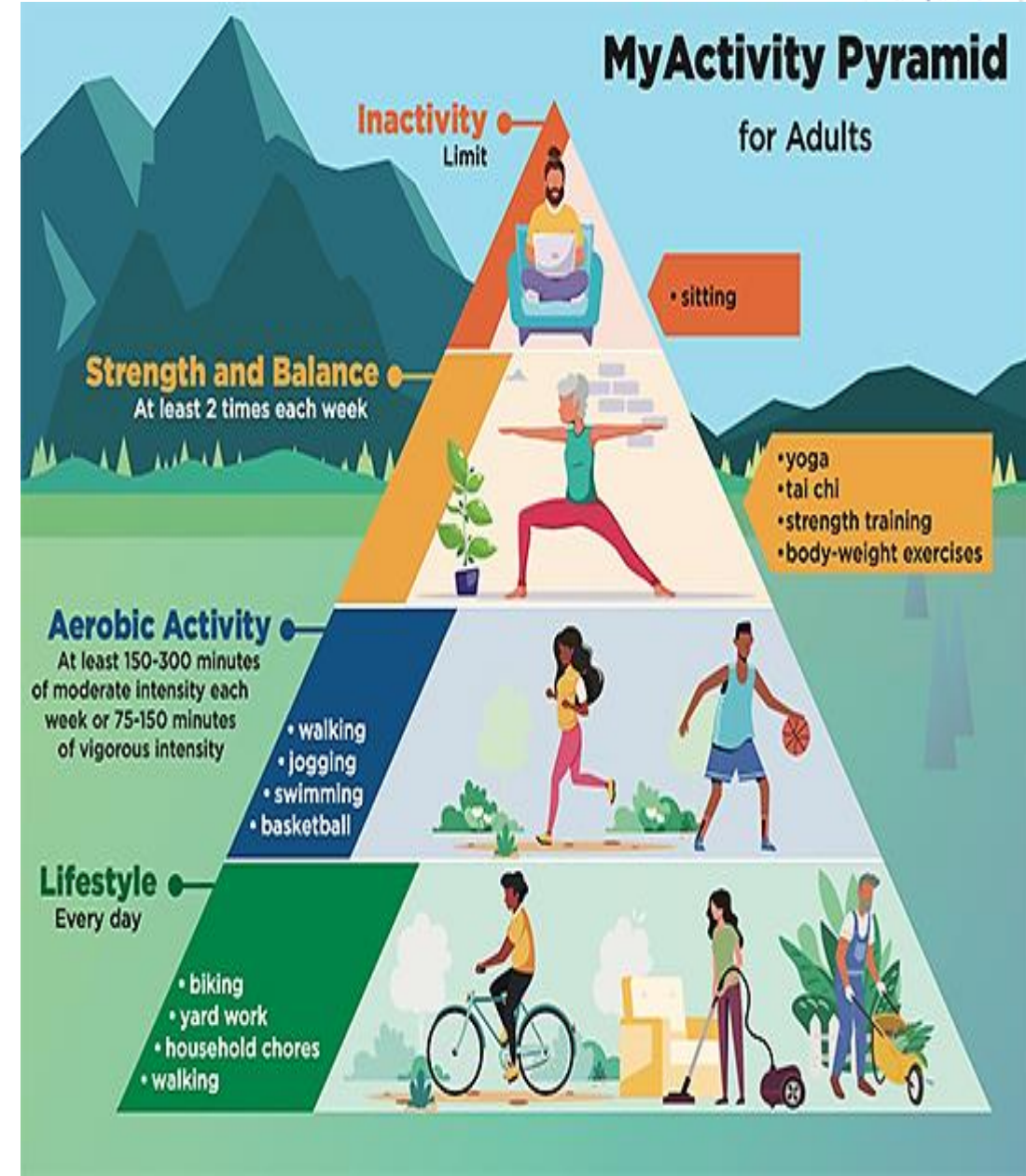


Physical activity

Aerobic exercise should be offered to patients with known CAD.

Patients with stable angina or stable chronic heart failure should undergo moderate-to-vigorous intensity aerobic exercise training ≥ 3 times a week and for 30 min per session.

Sedentary patients should be strongly encouraged to start light-intensity exercise programs.



TYPICAL EXERCISE SESSION

- Warm-up (five to ten minutes)
- Aerobic conditioning (up to 30 minutes or more depending upon the activity and exercise prescription) walking, jogging, cycling and climbing stairs are all excellent aerobic activities
- Cool-down (five to seven minutes)

Physical activity

- A large variation in physical activity insufficiency exists between regions. Hail (91%), Madinah (90%), and Qasim (88%) have the highest percentages, while Tabuk (64%) and Bahah (65%) have the lowest.

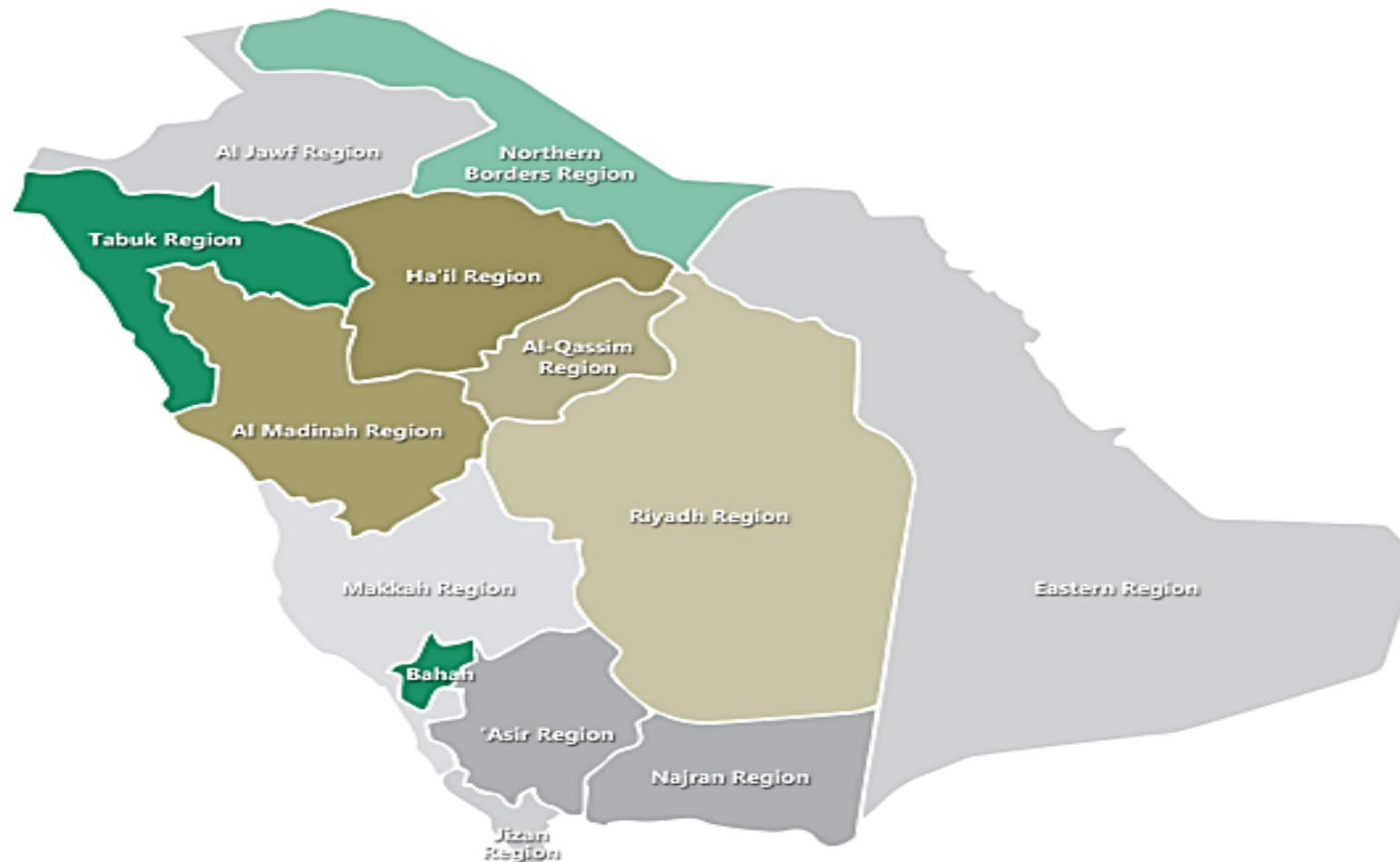
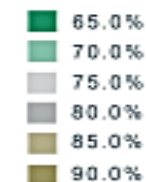


Figure 5.6: Physical activity by region

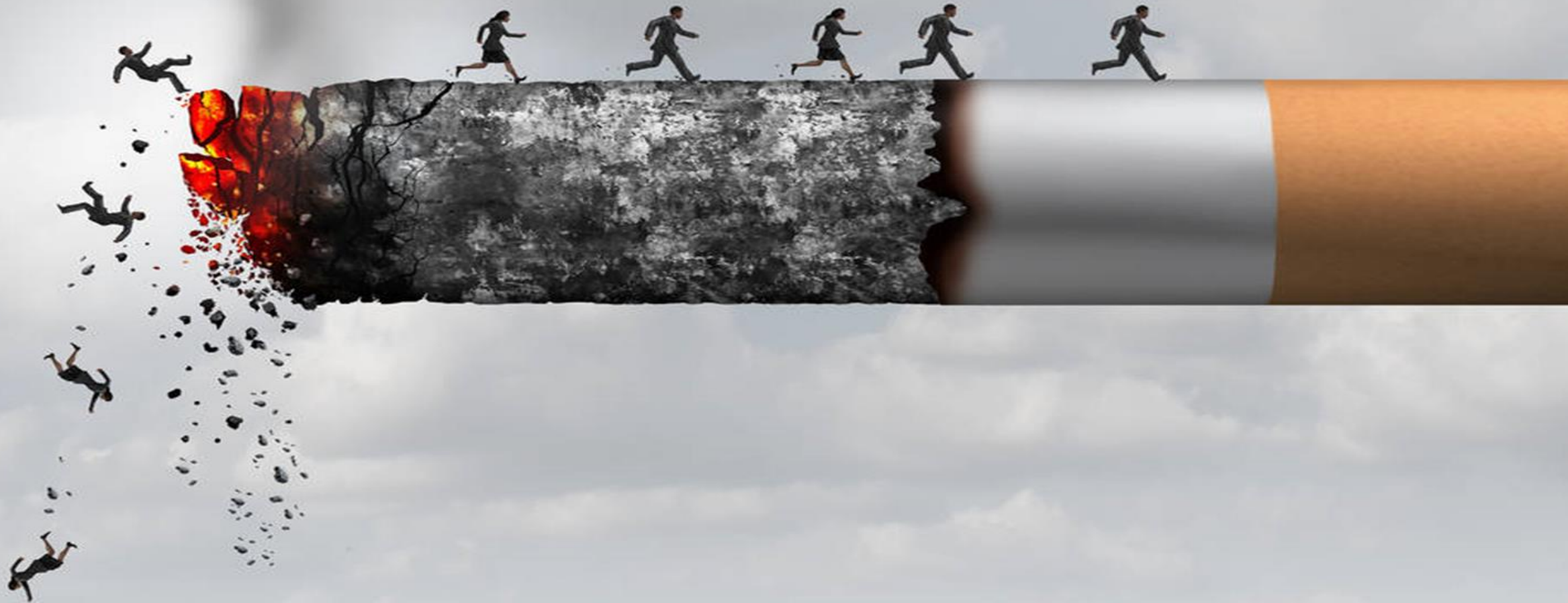
Percentage of respondents who have insufficient physical activity



Take home message is:

- Lifestyle improvement should begin early and underlie all preventive efforts
- Most cardiovascular events preventable with lifestyle modification
- Some is better than none, more is even better

Tobacco cessation



If You Quit Smoking Right now:



- Reduction in mortality of 36% after MI
- Nicotine replacement therapy is safe in patients with CAD and should routinely be offered.
- Bupropion and varenicline have been found safe to use in patients with stable CAD in some studies, although the safety of varenicline has been questioned in a meta-analysis, being associated with a small but statistically significant increase in CVD.

Hypertension

- Hypertension is the number one cardiovascular risk factor and the world's greatest risk factor for death and disability
- Prevalence estimated to be in the mid 20% range in Saudi
- People with HTN are three-times more likely to die from heart disease and four-times more likely to die from stroke

Hypertension

- Is home monitoring superior to in office monitoring?
- What is the optimal target for BP lowering during antihypertensive therapy in adults?
- In adults with hypertension, do different medication classes have better outcomes?
- In adults with hypertension, is it better to start with one drug or more than one class of drugs?

Hyperlipidemia

- Hyperlipidemia is one of the most important (and modifiable) risk factors for CAD
- It causes accelerated atherosclerosis
- 43% of Survey respondents in Saudi have raised serum cholesterol

Hyperlipidemia

Risk Factors

- Diet
- Age
- DM
- Inactive lifestyle
- Family history
- Gender
- Medications
- Genetic mutations

Hyperlipidemia

- **Fasting/non-fasting lipid profile:** Total cholesterol, HDL, & TGs
- Pathological values on two different occasions are required to confirm the diagnosis
- Dyslipidemia is diagnosed if LDL > 130 mg/dL and/or if HDL levels < 40 mg/dL.

Laboratory parameter	Optimal level	Pathological
Total Cholesterol	• 200 mg/dL	• Borderline: 200–239 mg/dL • High: > 240 mg/dL
TGs	• 150 mg/dL	• Borderline: 150–199 mg/dL • High: > 200 mg/dL • Very high: $\geq$ 500 mg/dL
LDL	• 100 mg/dL	• Near optimal: 100–129 mg/dL • Borderline high: > 130 mg/dL • High: > 160 mg/dL • Very high: $\geq$ 190 mg/dL
HDL	• $\geq$ 60 mg/dL	• Low: < 40 mg/dL
LDL/HDL ratio	• ♂: < 3.5 • ♀: < 3.0	• ♂: > 3.5 • ♀: > 3.0

Hyperlipidemia

Nonpharmacological measures:

- Dietary changes
- Reduce saturated fat and cholesterol intake
- Eliminate consumption of alcohol
- Maintaining a healthy weight
- Physical activity

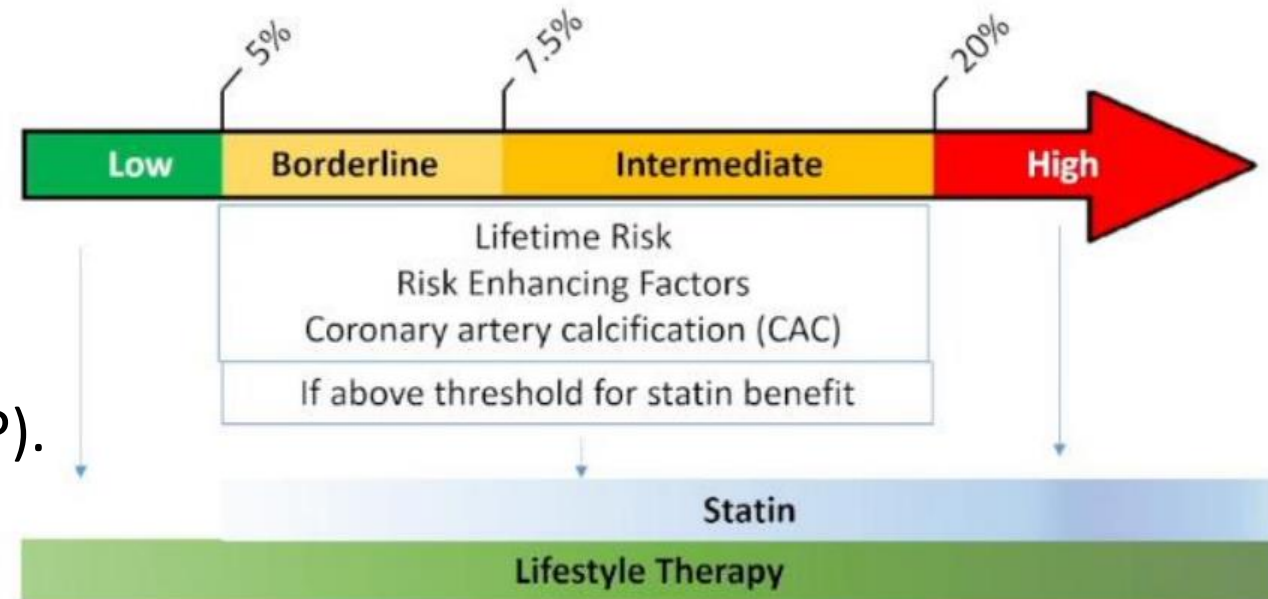
Medical therapy

- Statins
- Second-line lipid-lowering agents if there is a poor response to statins (PCSK9i)
(Proprotein convertase subtilisin/Kexin type 9 inhibitors)

Hyperlipidemia

Who Should Be Considered for Statins?

- Individuals with **clinical ASCVD** (atherosclerotic cardiovascular disease).
- Individuals with **LDL-c ≥ 190 mg/dL**, aged ≥ 21 years.
- **Diabetics** with LDL-c between **70-189 mg/dL**, aged **40-75 years**.
- Individuals with an **ASCVD risk $\geq 20\%$** .
- Individuals with an **ASCVD risk of 7.5%-20%** and **risk enhancers**, such as:
 - Family history of premature ASCVD.
 - Metabolic syndrome.
 - Chronic kidney disease (CKD).
 - Inflammatory conditions.
 - Elevated biomarkers (e.g., lipoprotein levels, high-sensitivity CRP).



Take home message is:

- First assess ASCVD risk
- Statins added to lifestyle to reduce risk of ASCVD in higher risk individuals and at maximally tolerated dose for highest risk patients (e.g. clinical ASCVD)
- PCSK9i reduce the risk of events when added to statin therapy in very high risk patients secondary prevention patients
- Lower is better

Diabetes Mellitus

- Diabetes is associated with a two- to three-fold increased risk of cardiovascular disease
- CVD is more likely to be the cause of hospitalization in patients with diabetes than diabetes-related causes
- Experts recommend prevention of cardiovascular events as the goal of diabetes treatment

Diabetes Mellitus

- Two new classes of drugs — sodium-glucose co-transporter 2 (SGLT-2) inhibitors and glucagon-like peptide-1 (GLP-1) receptor agonists showed a cardiovascular protection in dedicated outcome studies in patients with type 2 diabetes at high cardiovascular risk
- Dapagliflozin is an oral antidiabetic agent that can be added to the drug regimen of a patient with T2DM if monotherapy with metformin has not achieved the HbA1C target of $< 7\%$. This patient, however, has good glycemic control (HbA1C of 6.2%) on metformin alone and, thus, does not require a second antidiabetic drug.
- Should be considered for patients when atherosclerotic cardiovascular disease (ASCVD), heart failure, or chronic kidney disease predominates independent of HgbA1C

Diabetes Mellitus

The ADA recommends lifestyle management in addition to medications for the management of diabetes and cardiovascular risk factors, including nutrition therapy, physical activity and smoking cessation

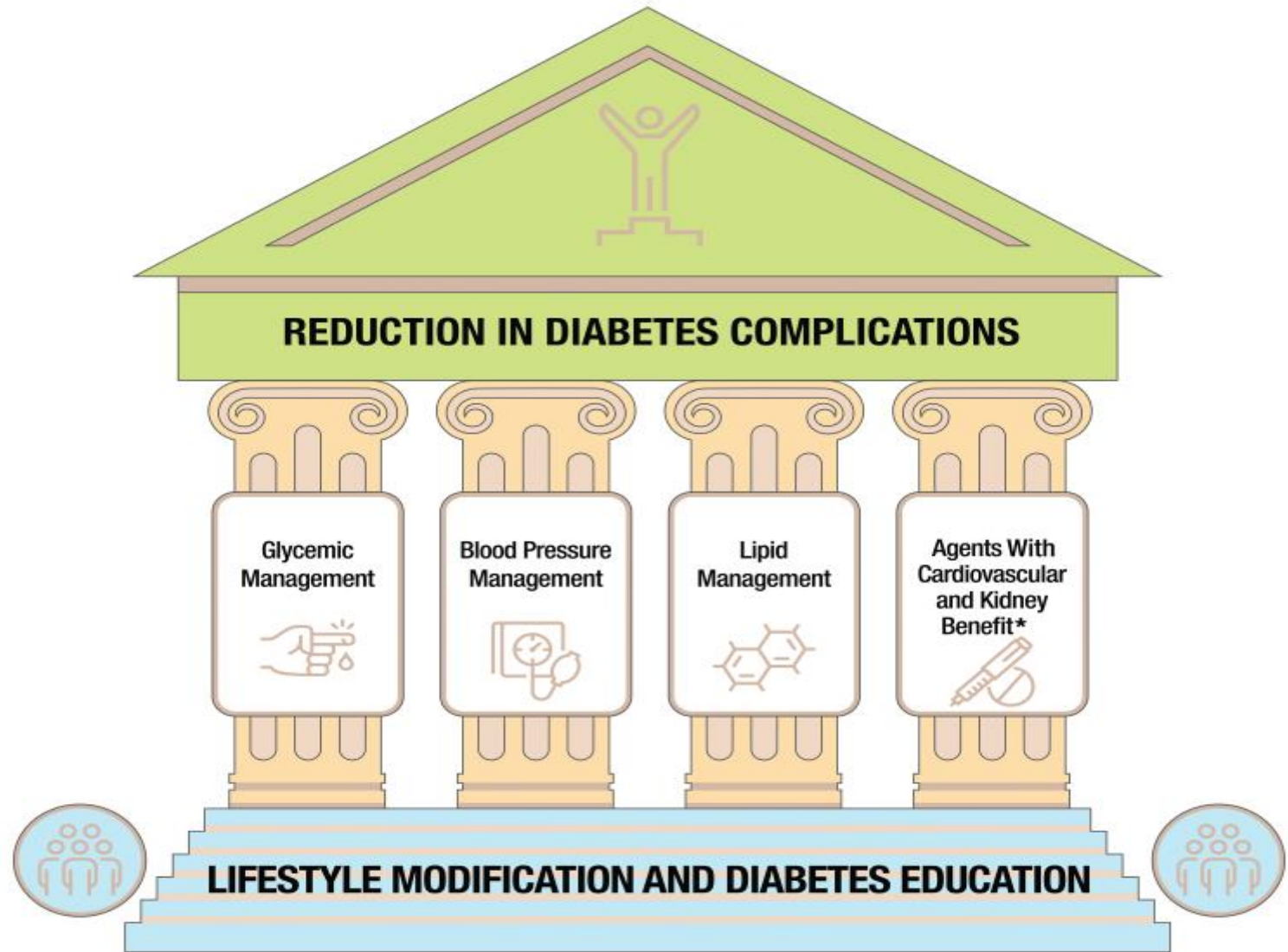


Figure 10.1—Multifactorial approach to reduction in risk of diabetes complications. *Risk reduction interventions to be applied as individually appropriate.

The U.S. Preventative Services Task Force (USPSTF) 2023

Population	Recommendation	Grade
Adults aged 40 to 59 years with a 10% or greater 10-year cardiovascular disease (CVD) risk	The decision to initiate low-dose aspirin use for the primary prevention of CVD in adults aged 40 to 59 years who have a 10% or greater 10-year CVD risk should be an individual one. Evidence indicates that the net benefit of aspirin use in this group is small. Persons who are not at increased risk for bleeding and are willing to take low-dose aspirin daily are more likely to benefit.	C
Adults 60 years or older	The USPSTF recommends against initiating low-dose aspirin use for the primary prevention of CVD in adults 60 years or older.	D

To Summarize

- Cardiovascular disease is a leading cause of death worldwide
- Assess your patients for cardiovascular risk
- Lifestyle improvement is the most important component of prevention and risk factor control (Life's simple 7)
- Statins added to lifestyle reduce risk in higher risk individuals, PCSK9i in the very highest risk patients
- Blood pressure control: target BP < 130/80 mm Hg for most patients
- Aspirin (low dose) in higher risk individuals if benefit outweighs risk of bleeding (Avoid in low risk individuals and elderly)

Case Scenario

A 54-year-old man comes to the office as a new patient. He feels well. He has type 2 diabetes mellitus treated with metformin. Routine colonoscopy screening from 2 months ago showed diverticulosis. His maternal aunt died from myocardial infarction at the age of 63 years. He does not smoke cigarettes and drinks three to four 12-oz beers weekly. He is 170 cm (5 ft 7 in) tall and weighs 70 kg (154 lb); BMI is 24 kg/m². His blood pressure is 124/68 mm Hg. Physical examination shows no abnormalities. Laboratory studies show:

Hemoglobin	15.2 g/dL
Leukocyte count	6000/mm ³
Hemoglobin A _{1c}	6.2%
Serum	
Total cholesterol	185 mg/dL
HDL cholesterol	45 mg/dL
LDL cholesterol	122 mg/dL
Triglycerides	140 mg/dL

The patient's estimated 10-year risk of cardiovascular disease is 9.1%.

Diabetes Mellitus

In addition to counseling on lifestyle modifications, which of the following health maintenance recommendations is most appropriate at this time?

- A. Aspirin therapy
- B. Atorvastatin therapy
- C. Dapagliflozin therapy
- D. No further action is indicated
- E. PCSK9i

References

- Step-up to medicine 5th edition
- Harrison's Principles of Internal Medicine 20th edition
- NEJM
- JACC
- AHA
- ADA
- Amboss



**If you want better,
go get better.**

Thank You

one-minute paper (2nd sem. 25-
26)

